

Instructions:

You will work on these exercises during the practical sessions in the fifth week of the course.

Below is an indication of the exercises that we will cover (at least) in each session. We encourage you try them before the session.

- First session (Wednesday/Thursday): exercises 1, 2.
- Second session (Friday): exercise 4.

Exercise 1: Constrained optimization

Consider the constrained optimization problem

$$\min_{\mathbf{x} \in \mathbb{R}^2} x_1 x_2 \text{ such that } x_1^2 + x_2^2 = 1.$$

Let's solve the problem via the method of Lagrange multipliers:

- (a) Write the constrained optimization problem in the form

$$\min_{\mathbf{x} \in \mathbb{R}^2} f(\mathbf{x}) \text{ subject to } g(\mathbf{x}) = c.$$

- (b) Write the Lagrangian $L(\mathbf{x}, \lambda)$.
- (c) Write down the stationary conditions for the Lagrangian.
- (d) Find all the stationary points $(\mathbf{x}^*, \lambda^*)$.
- (e) Which of the stationary points that you found are actually minimums?

Exercise 2: Mixture models

Consider a data distribution whose underlying generating process is a mixture of Poisson distributions, but we do not know the parameters of the mixture model. In this question, you are asked to derive the updated equations for the general Poisson mixture model. The Poisson distribution is:

$$p(x|\lambda) = \frac{1}{x!} \lambda^x \exp(-\lambda)$$

where $x = 0, 1, 2, \dots$ (non-negative integers), $\lambda > 0$ is the 'rate' of the data; the expected value of x is λ . A mixture representation assumes the following:

$$p(x_n) = \sum_{k=1}^K \pi_k p(x_n | \lambda_k)$$

where $p(x_n | \lambda_k)$ is a Poisson distribution with rate λ_k and x_n is a single data observation. To answer the following questions assume we are given a dataset $\{x_1, x_2, \dots, x_N\}$. Make sure that the constraint $\sum_k \pi_k = 1$ is satisfied (i.e. think of the log-likelihood or log-joint as f (an objective to maximize) and $\sum_k \pi_k - 1 = 0$ as $g = 0$ (a constraint that must hold)).

- (a) Write down the likelihood (as usual) for the data set in terms of $\{x_1, x_2, \dots, x_N\}$, $\{\pi_k\}$ and $\{\lambda_k\}$.
- (b) Write down the log-likelihood (as usual) for the data set in terms of $\{x_1, x_2, \dots, x_N\}$, $\{\pi_k\}$, $\{\lambda_k\}$.
- (c) Let us consider the contribution of each of the Poisson components as their *responsibility* in the generating process, denoting them r_{nk} (i.e. the contribution of the k th- poisson distribution for the n th datapoint).
This responsibilities are actually the posterior probabilities given by the following expression $p(C_k | \mathbf{x}_n) = r_{nk}$.
Find the expression for the responsibilities r_{nk} .
- (d) Find the expression for λ_k that maximizes the log-likelihood given fixed responsibilities $\{r_{nk}\}_{n=1}^N$.
- (e) Find the expression for π_k that maximizes the log-likelihood given fixed responsibilities $\{r_{nk}\}_{n=1}^N$.
- (f) Now assume priors for π_k and λ_k , $p(\lambda_k | a, b) = \mathcal{G}(\lambda_k | a, b)$ (a Gamma prior) and $p(\pi_1, \dots, \pi_K) = \mathcal{D}(\pi_1, \dots, \pi_K | \alpha/K, \dots, \alpha/K)$ (a Dirichlet distribution) respectively. These distributions are defined in the appendix of Bishop. Write down the log-joint distribution:

$$\log p(x_1, \dots, x_N, \{\pi_k\}, \{\lambda_k\} | a, b, \alpha, K).$$

- (g) Find the expression for λ_k that maximizes the log-joint given fixed responsibilities $\{r_{nk}\}_{n=1}^N$.
- (h) Find the expression for π_k that maximizes the log-joint given fixed responsibilities $\{r_{nk}\}_{n=1}^N$.
- (i) Write down an iterative algorithm using the above update equations (similar to the ones derived in class for the Mixture of Gaussians); include initialization and convergence check steps.

Exercise 3: Mixture of Experts in the Mushroom Kingdom

Exam question

Mamma mia! The evil Bowser has kidnapped Princess Peach again. Mario is about to start his journey to Bowser's castle, but, this time, he wants to apply some of the knowledge he acquired during the ML1 course to make his path a bit easier. More specifically, he wants to model the time between piranha plants coming out of their pipes to gain an advantage over them.

Mario has found some old files in Princess Peach's castle that state the following:

- There are three types of piranha plants: *regular*, *fire*, and *ice*.
- For a specific type of piranha plant, the time between pop-outs follows an exponential distribution, which has the following density:

$$p(x | \lambda) = \lambda e^{-\lambda x}, \quad x \geq 0, \quad \lambda > 0.$$

- He also found a data file with a list of pop-up times of piranha plants, but the plant types are not specified.

- (a) Given the information he gathered, *Mario* decided to model the time between appearances of piranha plants of any type with a mixture of exponential distributions.

Do/answer the following:

- (i) Write down the probability density function for such model.
 - (ii) Clearly define each of the random variables involved.
 - (iii) How many trainable parameters does the model have?
- (b) Write the log-likelihood function for a given dataset of samples $\mathcal{D} = \{x_1, x_2, \dots, x_N\}$ for this model.
- (c) Write down the expression for the responsibility of expert k on datapoint n , i.e., $r_{nk} = p(z = \mathcal{C}_k \mid x_n)$.

- (d) Mario now wants to apply the Expectation-Maximization algorithm to learn optimal parameters for the model. Formulate the optimization problems that need to be solved to update the parameters in the Maximization step.

Note: You do not need to compute the analytical expressions for the parameter updates, but only specify the optimization problem that needs to be solved to obtain them.

- (e) With his model trained, Mario enters a completely dark cave where he can't see anything. Suddenly, he hears a piranha plant nearby. By listening closely, he notices it pops out from its pipe every 5 seconds. How can he use his trained model to predict which type of piranha plant it is?

Note: you can assume that, after training the model, Mario has observed the resulting clusters and determined which one corresponds to each piranha plant type.

Exercise 4: Principal component analysis

Suppose we have a data set $\{\mathbf{x}_1, \dots, \mathbf{x}_N\}$ of D -dimensional vectors, which have a zero mean for each dimension. Assume we perform a complete eigenvalue decomposition of the empirical covariance matrix $\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$. You are interested in only a single projection of your data such that the variance of this projection is maximized. Let $\mathbf{u}_i$ be the direction vector of a particular projection. Assume that $\mathbf{u}_i^\top \mathbf{u}_i = 1$.

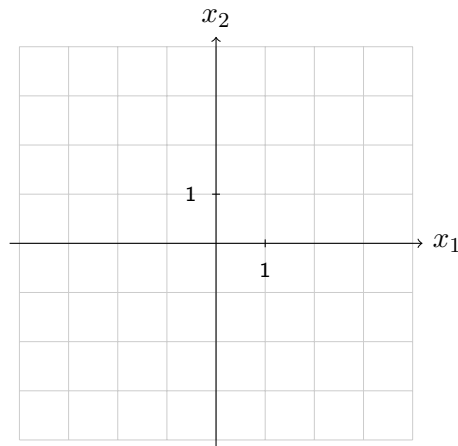
- (a) What is the projection z_{ni} of a given point $\mathbf{x}_n$ under the particular vector $\mathbf{u}_i$?
- (b) What is the empirical mean of the projection z_i across all points $\mathbf{x}_n$?
- (c) What is the empirical variance of the projection z_i ? Provide your answer in terms of the empirical covariance matrix $\mathbf{S}$
- (d) Replace $\mathbf{S}$ with its eigenvalue decomposition and simplify the aforementioned expression. What is the variance now?
with $\mathbf{e}_i$ to be a vector with zeros except the position with index i .
- (e) Suppose that you are interested in reducing the dimensionality from D to K , such that 99% of the variance is maintained. How can you select an appropriate K ?

Exercise 5: Dimensionality reduction

Exam question

We apply Principal Component Analysis (PCA) to a 2-dimensional dataset $\mathbf{x}$, obtaining that its first principal component is $\mathbf{u}_1 = \left(\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right)^\top$ and its second principal component is $\mathbf{u}_2 = \left(-\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right)^\top$.

- (a) Draw an example of a dataset $\mathbf{x}$ that satisfies this on the plot below. Include at least 10 samples.



For questions b and c we will assume that $\mathbf{x}$ is a dataset with zero mean.

- (b) Compute the projection of $\mathbf{x}_0 = (2, 2)^\top$ onto the subspace defined by the first principal component.
- (c) If we represent the projection of $\mathbf{x}_0 = (2, 2)^\top$ on the first component in the original 2D space, what will be the reconstruction error with respect to the original datapoint?